SQL AGGREGATION BASICS

Task 23 | Data Analytics Internship

Veda Technology

1. Objective

The objective of this task is to practise SQL aggregate functions and understand how they summarise data. The task focuses on COUNT, SUM, AVG, MIN and MAX, along with GROUP BY and how aggregate functions handle NULL values.

2. Tools and Dataset

Database: MySQL
SQL editor: MySQL Workbench
Dataset: Chinook sample database. Optional Northwind examples are included in the SQL script.

3. Aggregate Functions

Function	Purpose	Example use
COUNT()	Counts rows or non-NULL values in a column.	Number of invoices
SUM()	Adds numeric values.	Total invoice value
AVG()	Calculates the average of non-NULL numeric values.	Average invoice value
MIN()	Returns the smallest value.	Lowest invoice total
MAX()	Returns the largest value.	Highest invoice total
GROUP BY	Creates groups so aggregate values can be calculated per group.	Sales by billing country

4. Practice Queries

The SQL file contains 10 main queries covering the requested aggregate functions, plus additional GROUP BY practice. The examples below show the kind of analysis performed.

Count invoices

```
SELECT COUNT(*) AS total_invoices
FROM Invoice;
```

Calculate total invoice value

```
SELECT SUM(Total) AS total_invoice_value
FROM Invoice;
```

Find average invoice value

```
SELECT AVG(Total) AS average_invoice_value
FROM Invoice;
```

Find minimum and maximum invoice totals

```
SELECT MIN(Total) AS lowest_invoice_total,
MAX(Total) AS highest_invoice_total
FROM Invoice;
```

Summarise invoices by country

```
SELECT BillingCountry,
COUNT(*) AS invoice_count,
SUM(Total) AS total_sales,
AVG(Total) AS average_invoice_value
FROM Invoice
GROUP BY BillingCountry
ORDER BY total_sales DESC;
```

5. Understanding NULL Values

Aggregate functions generally ignore NULL values. COUNT(*) counts all rows, while COUNT(column) counts only rows where that column is not NULL. AVG() calculates the average using non-NULL values. SUM() and AVG() return NULL when there are no non-NULL values to process.

6. Interview Questions

Why use GROUP BY?

GROUP BY divides rows into groups, allowing aggregate functions such as COUNT, SUM and AVG to return a summary for each group.

How does AVG handle NULL values?

AVG ignores NULL values and calculates the average using only non-NULL values.

7. Conclusion

This task provides practice with the aggregate functions used to summarise database records. By combining these functions with GROUP BY, it becomes easier to compare categories, calculate totals and averages, and prepare data for analysis and reporting.

Prepared for Veda Technology | Task 23 – SQL Aggregation Basics